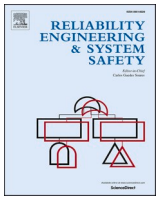
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Gradient-enhanced physics-informed long short-term memory networks for stable and accurate prediction of the RUL of electronic components

Chenyang Lai [a], Piero Baraldi [a,*], Enrico Zio [a,b]

[a] Energy Department, Politecnico di Milano, Milano, Italy
[b] MINES Paris-PSL University, CRC, Sophia Antipolis, France

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ABSTRACT

For effective maintenance decisions on electronic components, operators need predictions of the Remaining Useful Life (RUL) that are not only accurate but also stable. To this aim, a novel prognostic model based on gradient-enhanced Physics-Informed Neural Networks (gPINNs) is developed. It is based on mathematically formulating the physical fact that the ground-truth RUL of a component is reduced by one time unit for every unit of time elapsed in its life and its incorporation into the loss function of a Physics-Informed Long Short-Term Memory (PILSTM) network. To recover from possible inaccurate RUL predictions, which can occur especially at the end of the component life, a gradient-enhanced PILSTM is developed by considering the second derivative of the RUL, which should ideally be null. Additionally, a novel ensemble strategy is proposed for automatically weighing the different terms of the loss function so as to eliminate the labor-intensive and error-prone process of manually tuning the weights and to further improve the accuracy of the predictions. The proposed method is applied to two types of electronic components: Insulated Gate Bipolar Transistors (IGBTs) and lithium-ion batteries. The results demonstrate that it outperforms other state-of-the-art methods in terms of accuracy and stability of the RUL predictions.

1. Introduction

In recent years, several prognostics methods have been developed to support predictive maintenance of electronic components [1,2]. With the advancement of computational intelligence, Artificial Intelligence (AI)-based methods for predicting RUL of electronic components and systems are emerging. For example, a 1D Convolutional Neural Network (CNN) combined with an improved informer model was developed in [3] to predict the degradation evolution and RUL of supercapacitors in energy storage systems. An interpretable Long Short-Term Memory (LSTM) combined with an Autoencoder (AE) was developed in [4] to evaluate the degradation state of IGBTs. A conditional Graph Convolutional Network (GCN) was developed in [5] to predict the State of Health (SOH) and RUL of lithium-ion batteries. Beyond the field of electronic components, AI-based prognostic methods are also being explored for predictive maintenance in other industrial sectors. A multi-stage CNN combined with attention mechanisms was developed in [6] to predict the RUL of bearing and gear components. A dual-thread Gated Recurrent Unit (GRU) was developed in [7] to predict the RUL of gear components. Whereas existing prognostic methods primarily focus on the accuracy of the RUL predictions [8], additional attributes beyond accuracy must be considered to enhance the trustworthiness of the predictions for the maintenance-decision makers and, therefore, effectively assisting them in the maintenance decisions. Among these attributes, explainability [9], robustness to varying operational and environmental conditions [10], and stability [11] are the most important ones. Stable RUL predictions ensure that decisions are based on reliable trends rather than erratic outputs, fostering maintenance-decision makers trust in model outcomes. Also, stability contributes to the seamless integration of prognostic models into practical applications by reducing the risk of unnecessary maintenance interventions caused by oscillations of the RUL predictions. However, achieving stable RUL predictions for electronic components is challenged by: 1) competing failure modes, which complicate the degradation evolution due to their interactions [12], 2) variations of the operating conditions, which alter the degradation process, introducing large variability in the failure times [13]. 3) measurement errors and sensor malfunctions, which can compromise accuracy and stability of RUL predictions [14], 4) limited amount of run-to-failure data, which reduces the representativeness of the training set used for model development [15].

For example, Fig. 1 shows how a malfunctioning sensor can cause a

* Corresponding author.
E-mail address: piero.baraldi@polimi.it (P. Baraldi).

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Nomenclature

AAL	Average α-λ Accuracy
AAT	Accelerated Aging Test
ADI	Average Derivative Index
ASI	Average Steadiness Index
BiLSTM	Bidirectional Long Short-Term Memory
CRA	Cumulative Relative Accuracy
CNN	Convolutional Neural Network
DNN	Deep Neural Network
ESN	Echo State Network
gPILSTM	gradient-enhanced Physics-Informed Long Short-Term Memory
gPINN	gradient-enhanced Physics-Informed Neural Network
IGBT	Insulated Gate Bipolar Transistor
KF	Kalman Filter
ODE	Ordinary differential equation
PDE	Partial differential equation
PHM	Prognostics and Health Management
PILSTM-Tol	Physics-Informed Long Short-Term Memory based on Tolerance
RMSE	Root Mean Squared Error
RUL	Remaining Useful Life
SoH	State of Health

$x_t^{r,l}$	Value of signal l at time t of degradation trajectory r
t	Time
t_f^r	Failure time of degradation trajectory r
RUL_t^r	Ground-truth RUL at time t of degradation trajectory r
$\mathbf{h}_t$	Output of LSTM network at time t
L	Loss function
N_t	Number of training patterns
L_{data}	Data-driven term of the loss function
L_{phy}	Physical term of the loss function
L_{grad}	Gradient-enhanced term of the loss function
μ	Weight associated to L_{phy} in L
β	Weight associated to L_{grad} in L
$L_{phy-tol}$	Loss function used by *PILSTM-Tol*
σ	Tolerance used in $L_{phy-tol}$
G	Global performance indicator
R	Number of run-to-failure degradation trajectories
K	Number of disjoint subsets partitioned from run-to-failure degradation trajectories
$\widehat{RUL}^E$	RUL prediction of the Ensemble
$f_{lr}(\cdot)$	Cyclical learning rate function
$\overline{V}_{ce}$	Mean value of the collector-emitter voltage signal during each thermal cycle

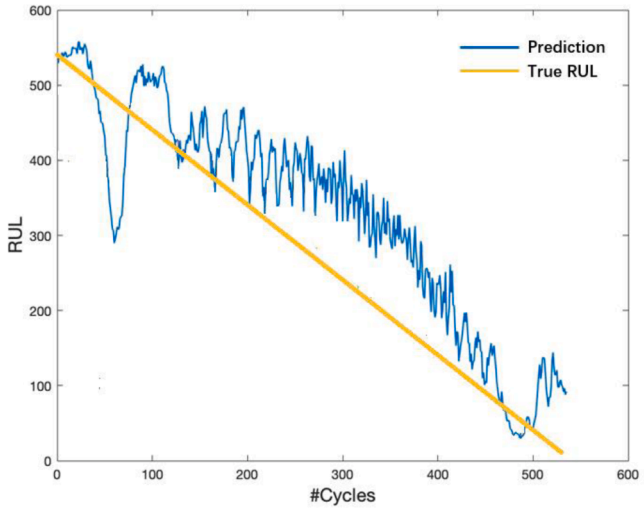


Fig. 1. Prediction of the RUL of a battery in case of a sensor malfunction (adapted from [16]).

deep learning model based on CNNs to produce inaccurate and unstable RUL predictions [16].

Whereas post-processing methods (reviewed in Section 2.1) can reduce RUL predictions instability through smoothing, they come at the cost of increasing model complexity and requiring parameter tuning, while not guaranteeing accuracy. Similar drawbacks exist in data pre-processing methods based on noise reduction [17], outlier removal and signal reconstruction.

Recently, Physics-Informed Neural Networks (PINNs) have been used for RUL prediction of industrial components. They were originally developed to solve partial differential equations (PDEs) by integrating physical constraints, initial or boundary conditions into the loss function of Neural Networks (NNs) [18].

Applications of PINNs to RUL prediction are discussed in detail in Section 2.2. Wang et al. [19] addressed the weighing of the different terms of the loss function by developing a learning rate annealing algorithm that utilizes gradient statistics to properly adjust the weights of the different terms. Yu et al. [20] developed a gPINN to improve the accuracy of a PINN, the idea is to add to the loss function an additional gradient-enhanced term, which aims at reducing the gradient of the PDE residual.

Inspired by gPINN, the present work develops a gradient-enhanced Physics-Informed Long Short-Term Memory (gPILSTM) network. LSTM cells are used to capture the nonlinear temporal behaviour of the degradation process [21]. The physical fact that the ground-truth RUL is reduced by one time unit for every unit of time elapsed, i.e.:

$$RUL(t) = RUL(t-1) - 1 \tag{1}$$

is incorporated in the loss function of the PINN to enhance the stability of RUL predictions. Specifically, we formulate Eq. (1) using the differential equation $\frac{dRUL}{dt} = -1$ and its gradient-enhanced term $\frac{d^2RUL}{dt^2} = 0$. These two differential equations are integrated into the loss function of gPILSTM to enhance accuracy and stability, without the need of additional post-processing.

Note that Eq. (1) has been employed in [22] to define the state transitional process within a semi-stochastic filtering approach, where the RUL of the component is its actual (unobservable) state. However, this approach requires a model describing the relationship between the RUL and the monitored signals, which is typically unavailable for electronic components.

The main contributions of the work are:

1) incorporation of the physical constraint $\frac{dRUL}{dt} = -1$ within a PILSTM approach to enhance the stability of RUL predictions. Unlike conventional approaches that apply post-processing techniques to smooth RUL predictions, the proposed method allows simplifying the overall prognostic pipeline and enhancing its stability by directly providing smooth predictions.
2) addition of a gradient-enhanced loss term incorporating the physical constraint $\frac{d^2RUL}{dt^2} = 0$ to the physical and data-driven losses. The gradient-enhanced loss was initially proposed in [20], to solve PDEs. However, its application to RUL prediction in a purely data-driven

regression context represents a novelty, which allows bridging the gap between physical consistency and data-driven modeling in prognostic applications.

3) a novel ensemble strategy for automatically weighing the different terms of the loss function. The use of multiple models eliminates the labor-intensive and error-prone process of manually tuning the weights, significantly increasing the ease of implementation and deployment of the prognostic approach.

The proposed method is verified in two prognostic case studies on two different electronic components. The former considers simulated run-to-failure trajectories of IGBTs of inverters used in fully electric vehicles, whereas the latter considers the experimental dataset of lithium-ion batteries cycled under fast-charging conditions [23].

The remainder of this paper is organized as follows. Section 2 provides an overview of related works. In Section 3, the theoretical basis of gPINNs is introduced. Section 4 details the problem addressed, while the proposed prognostic method based on gPILSTMs is presented in Section 5. In Section 6, two case studies on RUL prediction for IGBTs and lithium-ion batteries are presented, comparing the proposed method with state-of-the-art methods. Finally, Section 7 concludes the paper.

2. Related work

This section reviews existing techniques for RUL prediction smoothing (Section 2.1) and PINN-based methods for fault prognostics (Section 2.2).

2.1. Methods for smoothing RUL predictions

Wang et al. [24] developed a hybrid prognostic method using exponential degradation models coupled with the Frechet distance which improved the accuracy and the stability of RUL predictions. Shang et al. [25] used a weighted average of the RUL predicted within a time period to effectively reduce the prediction fluctuations. Ren et al. [26] built a linear regression model of the relationship between the time instants of prediction within a given time period and the RUL values correspondingly predicted; the regressor model was, then, used to predict the RUL at the end of the period of interest. Ren et al. [27] proposed to use linear filters based on least squares polynomial approximation to perform a posteriori smoothing of predicted RULs. Baraldi et al. [28] proposed a Kalman filter-inspired strategy for aggregating the outcomes of an ensemble of linear models. All these methods are based on post-processing RUL predictions, which adds complexity to the overall prognostic model and requires to set additional parameters while not necessarily guaranteeing the accuracy of final prediction. A completely different approach has been developed in [29] where a consistent, monotonic and gradually-decreasing-with-time health indicator was built and used to provide stability for RUL predictions. Zhang et al. [30] developed a NN whose loss function included a term for minimizing the non-linearity of RUL predictions in intervals of fixed length Δ. The proposed method was shown to improve the smoothness of the RUL predictions without manual post-processing, but requires to properly set a priori the parameter Δ on the basis of empirical knowledge from practical scenarios.

2.2. PINNs for fault prognostics

Some PINN-based methods for fault prognostic are already investigated [31]. He et al. [32] developed a physics-informed graph neural network to estimate the RUL of aeroengines. where the loss function was defined to penalize more overestimation than underestimation of the RUL. Yang et al. [33] developed a trainable discrete Fourier transform module which was integrated into a NN to adaptively extract frequency-related hidden features for the estimation of the RUL of rolling bearings. Huang et al. [34] developed a physics-informed deep autoencoder to extract health indicators for lithium-ion batteries and electromechanical actuators with the loss function penalizing the neural network parameters that lead to estimating the batteries healthier than in their initial state, which is unplausible. Zheng et al. [35] developed a PINN to predict crack growth of quasi-brittle materials under complex loads. In this case, the loss function included the energy variational principle and the boundary conditions as penalty terms, reducing the need of using historical degradation data for model training. Cofre-Martel et al. [36] developed a Deep Neural Network (DNN) was to map the monitored signals and operating time into a degradation indicator, then, the derivatives of the RUL with respect to operating time and the features extracted by the hidden layer of the DNN are formulated as a PDE, and eventually incorporated into the loss function.

3. Gradient-enhanced physics-informed neural network

This section briefly introduces PINNs and gPINNs for solving the PDE:

$$f\left(\boldsymbol{x};\frac{\partial y}{\partial x_1},\cdots,\frac{\partial y}{\partial x_d};\frac{\partial^2 y}{\partial x_1\partial x_1},\cdots,\frac{\partial^2 y}{\partial x_1\partial x_d};\ldots;\boldsymbol{\lambda}\right)=0,\ \boldsymbol{x}=(x_1,\cdots,x_d)\in\Omega \tag{2}$$

with boundary conditions

$$\mathscr{B}(y,\boldsymbol{x})=0 \text{ on } \partial\Omega \tag{3}$$

where $\boldsymbol{x}$ is a d-dimensional vector, defined on a domain Ω, representing the independent variables of the PDE, y is the dependent variable of the PDE, $\boldsymbol{\lambda}$ is the vector of parameters of the PDE and the boundary conditions $\mathscr{B}(\cdot)$ specify the solutions of the PDE within the boundaries $\partial\Omega$ of the domain Ω.

In all generality, a PINN is a neural network $\phi(\boldsymbol{x};\boldsymbol{\theta})$ that provides in output a reconstruction $\widehat{y}$ of y [18]. The constraints implied by the PDE $f(\cdot)$ (Eq. (2)), and the boundary conditions $\mathscr{B}(\cdot)$ (Eq. (3)) are integrated as a weighted sum into the loss function $\mathscr{L}$ of the neural network $\phi(\boldsymbol{x};\boldsymbol{\theta})$:

$$\mathscr{L}(\mathscr{T})=w_f\mathscr{L}_f(\mathscr{T}_f)+w_b\mathscr{L}_b(\mathscr{T}_b) \tag{4}$$

with loss terms:

$$\mathscr{L}_f(\mathscr{T}_f)=\frac{1}{N_{\mathscr{T}_f}}\sum_{\boldsymbol{x}\in\mathscr{T}_f}\left|f\left(\boldsymbol{x};\frac{\partial\widehat{y}}{\partial x_1},\ldots,\frac{\partial\widehat{y}}{\partial x_d};\frac{\partial^2\widehat{y}}{\partial x_1\partial x_1},\ldots,\frac{\partial^2\widehat{y}}{\partial x_1\partial x_d};\ldots;\boldsymbol{\lambda}\right)\right|^2 \tag{5}$$

$$\mathscr{L}_b(\mathscr{T}_b)=\frac{1}{N_{\mathscr{T}_b}}\sum_{\boldsymbol{x}\in\mathscr{T}_b}|\mathscr{B}(\widehat{y},\boldsymbol{x})|^2 \tag{6}$$

The loss term $\mathscr{L}_f$ evaluates the solution accuracy within the domain Ω and is computed on a set of $N_{\mathscr{T}_f}$ data patterns $\mathscr{T}_f$ selected inside the domain Ω; the loss term $\mathscr{L}_b$ evaluates the solution accuracy on the boundary of domain Ω and is computed using a set of $N_{\mathscr{T}_b}$ data patterns $\mathscr{T}_b$ selected on the boundary $\partial\Omega$.

As pointed out in [20], a PINN may exhibit unsatisfactory accuracy in the solution of PDEs, especially when only a small dataset is available for its training (small $N_{\mathscr{T}_f}$). To address this issue, Yu et al. [20] proposed to embed the gradient of PDE into the loss function (Eq. (4)). Specifically, since $f(\boldsymbol{x})$ is zero for any $\boldsymbol{x}\in\Omega$, also the derivatives of $f(\boldsymbol{x})$ are equal to zero. Then, the gradient of the PDE governing function $f(\cdot)$ to $\boldsymbol{x}$, $\nabla f(\boldsymbol{x})$, exists and is equal to zero:

$$\nabla f(\boldsymbol{x})=\left(\frac{\partial f}{\partial x_1},\frac{\partial f}{\partial x_2},\cdots,\frac{\partial f}{\partial x_d}\right)=0,\ \boldsymbol{x}\in\Omega \tag{7}$$

Then, the derivatives $\nabla f(\cdot)$ can be additionally integrated into the loss function $\mathscr{L}$, which becomes:

$$\mathscr{L}(\mathscr{T})=w_f\mathscr{L}_f(\mathscr{T}_f)+w_b\mathscr{L}_b(\mathscr{T}_b)+\sum_{i=1}^{d}w_{g_i}\mathscr{L}_{g_i}(\mathscr{T}_{g_i}) \tag{8}$$

with:

$$\mathscr{L}_{g_i}(\mathscr{T}_{g_i}) = \frac{1}{N_{\mathscr{T}_{g_i}}} \sum_{\boldsymbol{x} \in \mathscr{T}_{g_i}} \left| \frac{\partial f}{\partial x_i} \right|^2 \tag{9}$$

The loss term $\mathscr{L}_{g_i}$ evaluates the discrepancy between 0 and the gradients values and is computed using a set of $N_{\mathscr{T}_{g_i}}$ data patterns $\mathscr{T}_{g_i}$ typically set equal to $\mathscr{T}_f$. The general idea of using gradient information has also been demonstrated useful for function regression, e.g. Gaussian process [37] and neural networks [38].

4. Problem setting

We assume to have available R run-to-failure degradation trajectories of similar electronic components, which are assumed to cover all operating conditions. The generic rth trajectory, $X^r_{1:t^r_f} = \left[\mathbf{x}^{r,l}_{1:t^r_f}\right]_{l=1\ldots L} \in \mathbb{R}^{L \times t^r_f}$, $r = 1, \ldots, R$, consists of the time series of L signals, $\mathbf{x}^{r,l}_{1:t^r_f} = \left[x^{r,l}_t\right]_{t=1\ldots t^r_f} \in \mathbb{R}^{t^r_f}$, $l = 1, \ldots, L$, measured from the onset of the degradation until the component failure time t^r_f, i.e. the time at which the component fails to perform its intended function [39].

We consider the problem of predicting the RUL of a test component at the current time t, i.e. the amount of time the component can continue performing its function [10]:

$$RUL_t = t_f - t \tag{10}$$

The problem is addressed by developing a prognostic model, which receives as inputs the signal measurements $X^{test}_{1:t}$, i.e. the signal values collected from the degradation onset until the present time t, and provides in output the prediction $\widehat{RUL}_t$ of RUL_t. This work aims to develop a prognostic model that simultaneously minimizes the error in the RUL prediction $\widehat{RUL}_t$ and maximizes the stability of the sequence of RUL predictions, $\widehat{RUL}_{1:t_f}$, throughout the operational life of electronic components. We assume that the test components are of the same type of those used to generate the training data and that they operate under similar conditions.

With regard to the minimization of RUL prediction error, *Cumulative Relative Accuracy (CRA), Average $\alpha - \lambda$ Accuracy (AAL), Root Mean Squared Error (RMSE)* are considered as metrics to evaluate the model performance [40]. Appendix 1 reports their definitions.

With regard to the evaluation of the stability of RUL prediction, the metric *Average Steadiness Index (ASI)* has been proposed in [40]. Given a set of R^{test} components, *ASI* considers the standard deviation of the error, $d^r(t) = RUL^r_t - \widehat{RUL}^r_t$, between the predicted and ground-truth RUL over a sliding time window of *tw* time steps. Specifically, the obtained standard deviation is averaged along the entire lifetime of the component:

$$SI^r = \frac{1}{t^r_f - tw + 1} \sum_{t=1}^{t^r_f - tw + 1} std([d^r(t), d^r(t+1), \cdots, d^r(t+tw-1)]) \tag{11}$$

and over all test components:

$$ASI = \frac{1}{R^{test}} \sum_{r=1}^{R^{test}} SI^r \tag{12}$$

The smaller is the ASI value, the more stable are the RUL predictions.

A limitation of this metric is that its value is remarkably influenced by the length of the time window *tw*, over which the standard deviation of the RUL prediction error is computed. For this reason, we propose a new metric, *Average Derivative Index (ADI)*, which does not require to consider a time window and is directly inspired by the principle that each interval of time that elapses between two predictions should reduce the predictions by the same interval. Assuming that the RUL is predicted at time intervals of one time unit, the derivative index, DI^r, for test component r is:

$$DI^r = \frac{1}{t^r_f - 1} \sum_{t=1}^{t^r_f - 1} \left| \widehat{RUL}^r_t - \widehat{RUL}^r_{t-1} + 1 \right| \tag{13}$$

Then, the average DI^r over all R^{test} trajectories is computed as:

$$ADI = \frac{1}{R^{test}} \sum_{r=1}^{R^{test}} DI^r \tag{14}$$

The smaller is the *ADI* value, the smoother are the RUL predictions.

5. Method

The proposed method is based on an ensemble of gradient-enhanced Physics-informed Long Short-Term Memory networks (gPILSTM). Sections 5.1 and 5.2 will describe the two main contributions of the proposed method: (1) the integration of the physical constraints into the LSTM model, and (2) an ensemble strategy to appropriately assign the weights of the different terms of the PINN loss function.

5.1. Gradient-enhanced physics-informed long short-term memory networks

The prognostic method proposed for RUL prediction is based on the use of LSTM cells. The choice is motivated by the sequential nature of sensor measurements $X^{test}_{1:t}$ and the nonlinear behaviour of the degradation process of many types of components, which make LSTMs well-suited to predict the component RUL_t from time-series data [41]. More details on LSTM are given in Appendix 2.

The LSTM network, formulated as $\widehat{RUL}_t = \phi([X_{1:t}, t]; \boldsymbol{\theta}_L)$, receives in input the time series of the L measured signals ($X_{1:t}$) and the time elapsed from the onset of the degradation process (t) to provide in output the estimation of $\widehat{RUL}_t$ of the RUL. The loss function used for model training contains a data-driven term:

$$L_{data} = \frac{1}{N_t} \sum_{r=1}^{R} \sum_{t=1}^{t^r_f} \| RUL^r_t - \widehat{RUL}^r_t \|^2 \tag{15}$$

to exploit the available run-to-failure trajectories $\left\{X^r_{1:t^r_f}, RUL^r_{1:t^r_f}\right\}_{r=1}^{R}$.

To enhance further the stability of the RUL prediction, the physical fact that the ground-truth RUL is reduced by one time unit for every unit of time elapsed (Eq. (1)) is incorporated into the LSTM network, resulting in a gPILSTM. Specifically, Eq. (1) is formulated using the ordinary differential equation (ODE):

$$\frac{dRUL}{dt} + 1 = 0 \tag{16}$$

which is a special case of the PDE in Eq. (2) (Section 3), where *RUL* and t correspond to y and $\boldsymbol{x}$, respectively. Furthermore, differentiating Eq. (16) yields:

$$\frac{d^2RUL}{dt^2} = 0 \tag{17}$$

which is analogous to Eq. (7) of the PDE.

The term L_f of the PINN loss function (Eq. (8)), which accounts for the solution accuracy within the domain, becomes:

$$L_f = \frac{1}{N_t} \sum_{r=1}^{R} \sum_{t=1}^{t^r_f} \| \frac{d\widehat{RUL}^r_t}{dt} + 1 \|^2 \tag{18}$$

with $N_t = \sum_{r=1}^{R} t^r_f$.

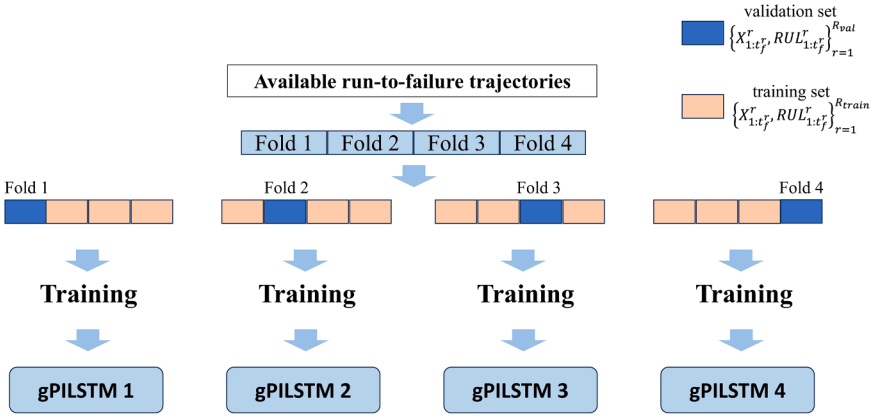
Similarly, the term $\mathscr{L}_g$ of the loss function (Eq. (8)), it becomes:

Algorithm 1
Procedure for training the gPILSTM.

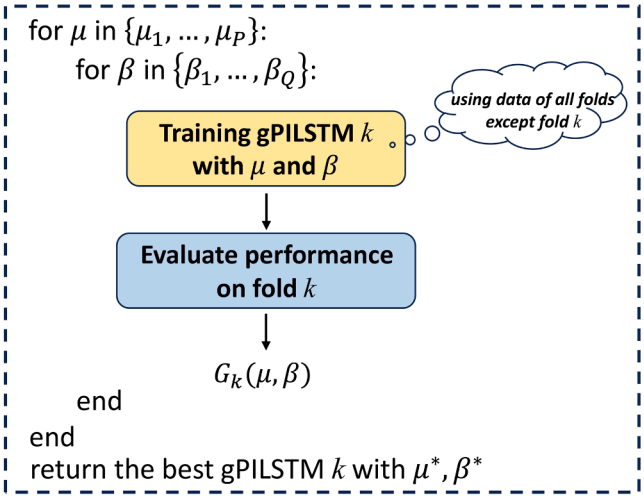
Input: training dataset: $D = \left\{X^r_{1:t^r_f}, RUL^r_{1:t^r_f}\right\}_{r=1}^{R}$; batch size: b; epoch number: E; weights of the physical loss terms: μ, β; learning rate: lr.
Output: optimal gPILSTM parameters $\boldsymbol{\theta}_L^*$.
1. initialize the parameters $\boldsymbol{\theta}_L$ of gPILSTM;
2. **for** $e = 1, \ldots, E$ **do**
3. build batch set $\left\{X^r_{1:t^r_f}, RUL^r_{1:t^r_f}\right\}_{r=1}^{b}$ from D;
4. update $\boldsymbol{\theta}_L$ by:

$$\boldsymbol{\theta}_L \leftarrow \boldsymbol{\theta}_L - lr\left\{(1-\mu-\beta)\frac{\partial L_{data}}{\partial \boldsymbol{\theta}_L} + \mu\frac{\partial L_{phy}}{\partial \boldsymbol{\theta}_L} + \beta\frac{\partial L_{grad}}{\partial \boldsymbol{\theta}_L}\right\}$$

5. **end**
6. $\boldsymbol{\theta}_L^* \leftarrow \boldsymbol{\theta}_L$
7. **return** $\boldsymbol{\theta}_L^*$



(a)



(b)

Fig. 2. (a) Workflow for the development of the ensemble of gPILSTM models; (b) Procedure for training the gPILSTM k.

$$L_{grad} = \frac{1}{N_t}\sum_{r=1}^{R}\sum_{t=1}^{t^r_f}\left\|\frac{d^2\widehat{RUL}^r_t}{dt^2}\right\|^2 \tag{19}$$

Since in this work no imposed boundary condition is on the ODE of Eq. (16), the corresponding loss term, $\mathscr{L}_g$, is omitted.

Eventually, the total loss function is given by the weighted sum of the terms in Eq. (15), Eq. (18) and Eq. (19):

$$L = (1-\mu-\beta)L_{data} + \mu L_{phy} + \beta L_{grad} \tag{20}$$

in which the hyperparameters μ and β indicate the relative importance of the physical loss term and the gradient-enhanced term with respect to L_{data}, respectively. Algorithm 1 reports the training procedure for the proposed gPILSTM.

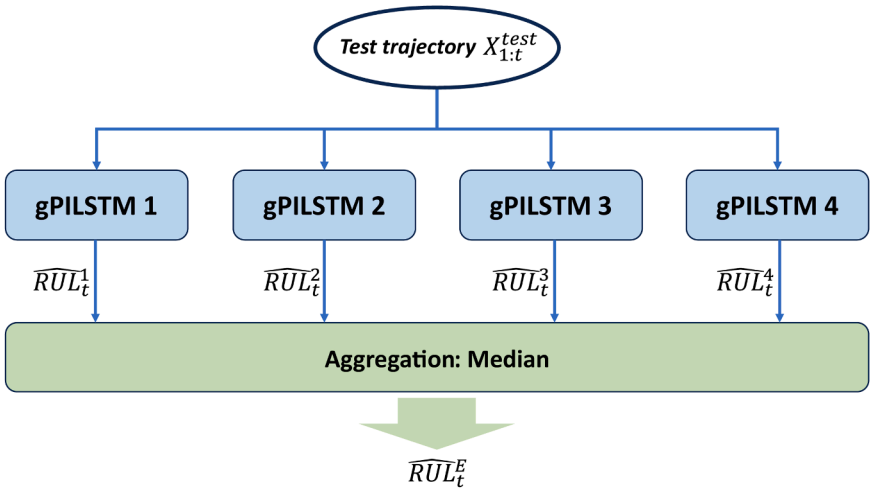


Fig. 3. Procedure for the prediction of the RUL of a test component at time t.

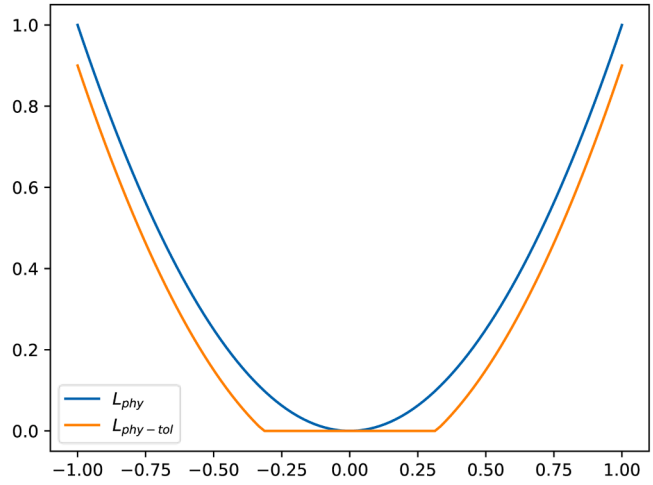


Fig. 4. Comparison between $L_{phy-tol}$ with $\sigma = 0.1$ and L_{phy}, as a function of $\frac{d\widehat{RUL}}{dt} + 1$.

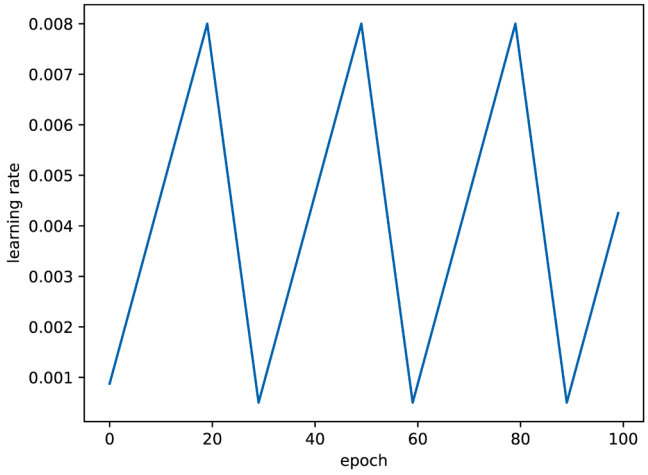


Fig. 5. Cyclical learning rate $f_{lr}(\cdot)$.

Table 1
LSTM architecture.

Parameters	Value
Number of hidden layers of the LSTM	2
Number of hidden states of the LSTM cell	16
Number of hidden layers of the FC layers	3
Number of neurons of the FC layers	[17,9,1]
Activation function of the LSTM and FC layers	Tanh()

Table 2
Training parameters.

Parameter	Value
Batch size	16
Optimizer	Adam
Epoch number	100
Learning rate	$f_{lr}(\cdot)$

5.2. An ensemble strategy for setting the weights

It has been observed that the training of PINNs largely depends on the values of the weights μ and β in the loss function (Eq. (20)). The difficulty is to identify combinations of μ and β that allow the minimization of the three terms of the loss function, avoiding the dominance of one over the others. A possible solution is to optimize the values of the hyperparameters μ and β using a validation set $\left\{X_{1:t_f^r}^r, RUL_{1:t_f^r}^r\right\}_{r=1}^{R_{val}}$ of R_{val} run-to-failure degradation trajectories extracted from the set of run-to-failure degradation trajectories $\left\{X_{1:t_f^r}^r, RUL_{1:t_f^r}^r\right\}_{r=1}^{R}$ available to develop the model. Specifically, a trial-and-error approach can be performed, where possible combinations of values of μ in the set $\{\mu_1,\ldots,\mu_P\}$, and β in the set $\{\beta_1,\ldots,\beta_Q\}$, with P and $Q > 1$, are considered. For each combination $\left[\mu_p, \beta_q\right]$, $p = 1,\ldots,P$, $q = 1,\ldots,Q$, a gPILSTM model is trained using the data $\left\{X_{1:t_f^r}^r, RUL_{1:t_f^r}^r\right\}_{r=1}^{R}$ and its performance is computed on the validation set $\left\{X_{1:t_f^r}^r, RUL_{1:t_f^r}^r\right\}_{r=1}^{R_{val}}$. In this work, since the final objectives are the accuracy and stability of the model, the hyperparameter setting $[\mu^*, \beta^*]$ which minimizes a weighted combination of the metrics [*CRA, AAL, RMSE, ASI, ADI*] defined in Section 2 is considered:

$$G_{p,q} = \sum_{m=1}^{M} w_m e_{m,norm}^{p,q} \tag{21}$$

where w_m is the weight assigned to the mth metric to reflect its importance. The negative values of *RMSE, ASI* and *ADI* are considered because

Table 3
CNN architecture used in IGBT case study.

Layer	Input Channels	Output channels	Kernel size	Padding size	Stride size	Input size	Output size
1st Convolutional layer	1	4	3	1	–	1 × 5	4 × 5
1st FC layer	–	–	–	–	–	20×1	8
2nd FC layer	–	–	–	–	–	8	1

Table 4
CNN architecture used in Battery case study.

Layer	Input channels	Output channels	Kernel size	Padding size	Stride size	Input size	Output size
1st Convolutional layer	23	32	3	1	–	23×50	32×50
1st Max pooling layer	–	–	2	–	2	32×50	32×25
2nd Convolutional layer	32	64	3	1	–	32×25	64×25
2nd Max pooling layer	–	–	2	–	2	64×25	64×12
1st FC layer	–	–	–	–	–	64×12	128
2nd FC layer	–	–	–	–	–	128	1

Table 5
Sets of values of the hyperparameters considered for the proposed and comparison methods.

Methods	Parameter	Candidates set
PILSTM	μ	[0.004,0.008,0.012,0.016,0.02,0.024,0.028,0.032,0.036,0.04]
PILSTM-Tol	μ	[0.008,0.016,0.024,0.032,0.04]
	σ	[0.1, 0.2,0.3,0.4, 0.5, 0.6]
gPILSTM	μ	[0.008,0.016,0.024,0.032,0.04]
	β	[0.1, 0.2,0.3,0.4, 0.5, 0.6]

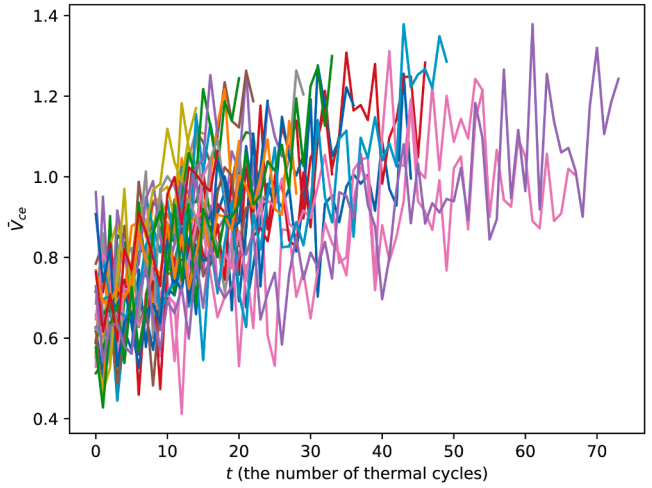


Fig. 6. $\overline{V}_{ce}$ evolution in run-to-failure trajectories.

Table 6
Performance of RUL predictions of the ablation study (IGBT case). The values in bold indicate the best performance among the methods.

Metrics	gPILSTM (proposed)	PILSTM	LSTM
CRA	**0.653±0.049**	0.611±0.030	0.646±0.049
AAL	0.625±0.049	**0.644±0.030**	0.603±0.053
RMSE	3.782±0.396	**3.686±0.351**	3.962±0.431
ASI	**0.662±0.037**	0.738±0.032	0.820±0.049
ADI	**0.604±0.027**	0.693±0.029	0.730±0.044
G	**0.901±0.057**	0.833±0.048	0.802±0.052

for these metrics, smaller is the value, more satisfactory is the performance. Also, the metrics are min-max normalized to $e^{p,q}_{m,norm}$ in the range $[0,1]$ to facilitate the assignments of the weights w_m. As a result, a dimensionless global performance indicator G is constructed. The larger is the G value, the better is the global performance.

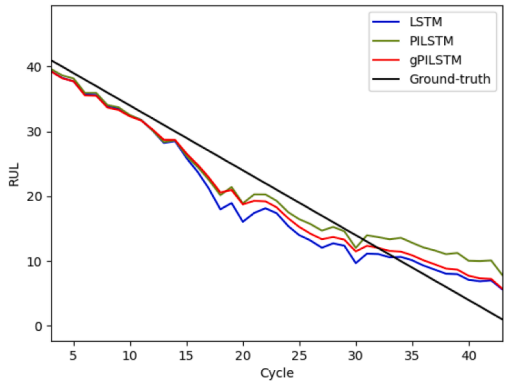
A difficulty related to hyperparameter setting in many industrial applications of fault prognostics is the limited number R of run-to-failure degradation trajectories available to develop the model. As a consequence, the validation set is made by a number $R_{val} < R$ of run-to-failure degradation trajectories that can be very small, with the risk that the setting $[\mu^*, \beta^*]$ of the hyperparameters identified by performing the above described trial-and-error procedure is optimal only for the specific validation set, whereas it does not provide satisfactory performances in a general case.

To overcome this problem, the present work develops an ensemble of gPILSTMs, where the hyperparameters $[\mu^*, \beta^*]$ of each model of the ensemble are optimized by trial-and-error using a different validation set. Specifically, the R run-to-failure degradation trajectories are randomly partitioned into K disjoint subsets of approximately equal size. Then, the kth subset is used as the validation set for the evaluation of the performance metrics whereas the remaining $K-1$ subsets are used to train the gPILSTM model and the trial-and-error procedure above described for the identification of the optimal hyperparameter setting $[\mu^*_k, \beta^*_k]$ is performed. The procedure is repeated until all K folds have served as validation set, so as to obtain an ensemble of K models $\{f_k(\cdot), k = 1, \ldots, K\}$, each one characterized by its tailored setting of the hyperparameters $[\mu^*_k, \beta^*_k]$. Fig. 2 illustrates the scheme of building the ensemble of gPILSTM models the case of $K = 4$, showing the process of training multiple gPILSTM models and combining their predictions.

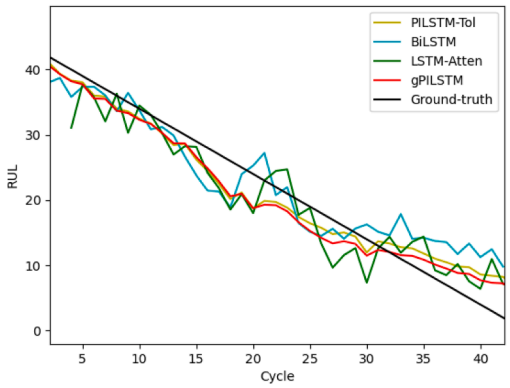
During the test phase, the signal measurements $X^{test}_{1:t}$ collected form the test component until the present time are fed as input to each kth model of the ensemble, $k = 1, \ldots, K$, which provides as output the RUL prediction $\widehat{RUL}^k_t = f_k(X^{test}_{1:t})$. Then, the RUL prediction of the ensemble, $\widehat{RUL}^E_t$, is computed by aggregating the K predictions $\widehat{RUL}^k_t$ of the K models using the median operator. This choice is motivated by the superior robustness of the median to possible outlier predictions compared to other possible operators, such as the mean [42]. Fig. 3 shows the

Table 7
Performance of RUL predictions compared with state-of-the-art methods (IGBT case). The values in bold indicate the best performance among the methods.

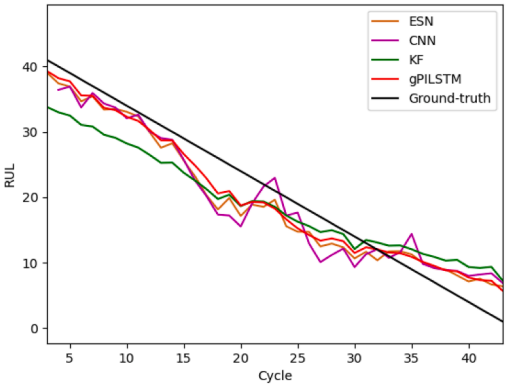
Metrics	gPILSTM (proposed)	PILSTM-Tol	BiLSTM	LSTM-Atten	ESN	CNN	KF
CRA	0.653±0.049	0.634±0.039	0.618±0.051	**0.667±0.022**	0.636±0.045	0.550±0.062	0.549±0.065
AAL	0.625±0.049	**0.637±0.054**	0.581±0.021	0.484±0.045	0.598±0.035	0.504±0.032	0.435±0.043
RMSE	3.782±0.396	**3.743±0.519**	5.049±0.350	5.660±0.419	4.363±0.691	5.081±0.324	6.352±0.458
ASI	**0.662±0.037**	0.700±0.034	1.384±0.108	1.527±0.142	0.803±0.052	1.299±0.094	0.683±0.041
ADI	0.604±0.027	0.644±0.026	1.406±0.117	1.582±0.175	0.766±0.065	1.200±0.085	**0.545±0.018**
G	**0.901±0.057**	0.875±0.063	0.400±0.089	0.240±0.094	0.746±0.116	0.266±0.073	0.454±0.064



(a) comparison within the ablation study.



(b) gPILSTM compared to LSTM based state-of-the-art methods.



(c) gPILSTM compared to non-LSTM based state-of-the-art methods.

Fig. 7. Predictions of the RUL during a run-to-failure trajectory.

procedure of the test phase.

6. Case studies

6.1. Methods considered for comparison

The proposed method is analyzed through an ablation study and by comparison with alternative state-of-the-art methods.

Concerning the ablation study, it is conducted by systematically removing key components of the proposed method. Specifically, we have considered:

- PILSTM - A PILSTM whose loss function $L = (1-\mu)L_{data} + \mu L_{phy}$ differs from the one of the proposed method due to the exclusion of the gradient enhanced term (L_{grad}). This comparison allows verifying the effectiveness of adding the term L_{grad} in the loss function.
- LSTM - A standard LSTM with loss function $L = L_{data}$ (Eq. (15)), which differs from the proposed method due to the exclusion of L_{phy} and L_{grad} from the loss function. This comparison allows verifying the effectiveness of using a PILSTM instead of a traditional LSTM.

Concerning the comparison with alternative state-of-the-art methods, we have considered methods recently proposed for time series analysis, namely:

- BiLSTM - A bidirectional LSTM with loss function $L = L_{data}$ (Eq. (15)) [43].
- LSTM-Atten – A standard LSTM aided by an attention mechanism with loss function $L = L_{data}$ (Eq. (15)) [44].
- PILSTM-Tol - A model belonging to the family of the PINN-Tols, which are state-of-the-art PINNs that introduce in the loss function of a PINN a new term to allow small deviations from the imposed physical laws [45,46], i.e.:

$$L = (1-\mu)L_{data} + \mu L_{phy-tol} \tag{22}$$

with

$$L_{phy-tol} = \frac{1}{N_t}\sum_{i=1}^{N_t} Relu\left(\left(\frac{d\widehat{RUL}_i}{dt}+1\right)^2 - \sigma\right) \tag{23}$$

where σ is the hyperparameter which controls the tolerance and $Relu(x) = \max(x, 0)$. The idea of PILSTM-Tol is to guarantee a tolerance to deviations caused by process or measurement noise. Fig. 4 shows the effect on the loss function L_{phy} when tolerance is considered.

Furthermore, we have considered for the comparison:

- an Echo State Network (ESN), due to its ability to model long-term dependencies in time-series.
- a 2D Convolutional Neural Network (CNN) processing a time window of the time series, due to its ability to capture short-term temporal dynamics.
- a Kalman Filter (KF), due to its ability to filter noise in the RUL prediction. Specifically, the RUL serves as state variable and the physical constraint (Eq. (1)) defines the state transition equation.

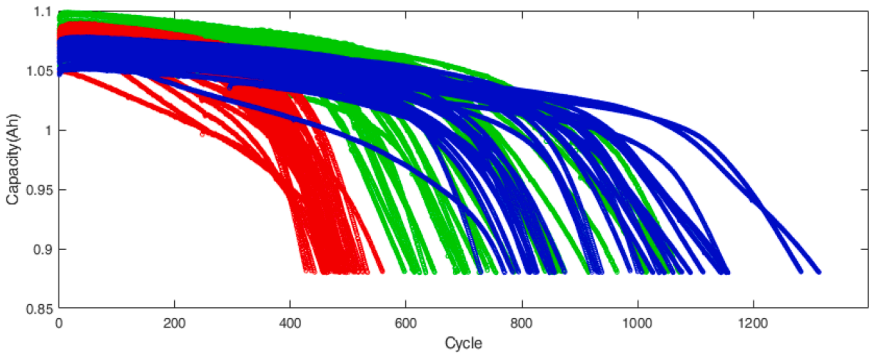


Fig. 8. Evolution of discharge capacity of the batteries. The three different colors refer to three batches of experiments characterized by different fast-charging conditions.

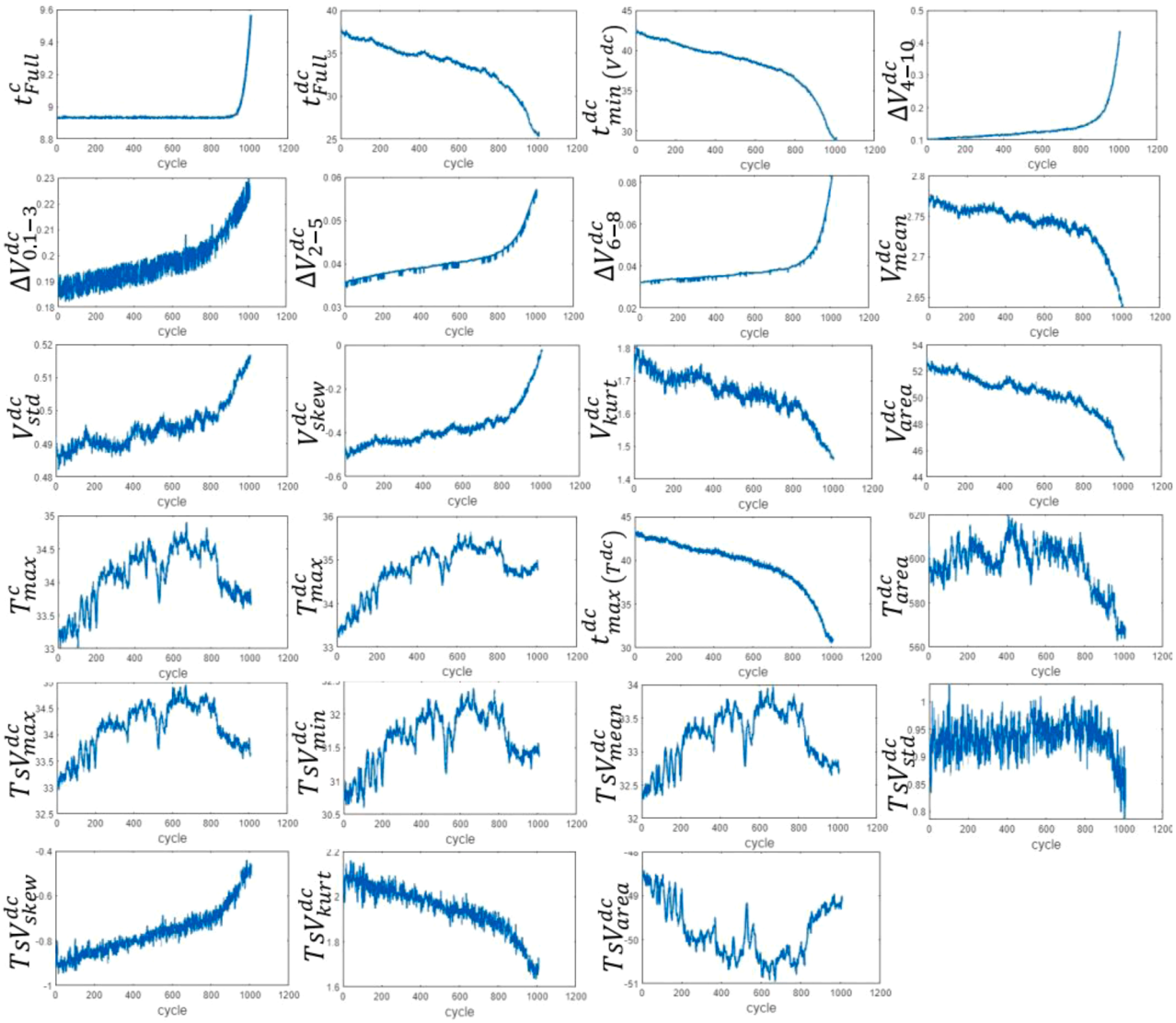


Fig. 9. Example of evolutions of the 23 features extracted during the life of a battery.

Since direct RUL observations are not available, a Linear Regression (LR) model mapping the input signals to RUL is built [28].

For fairness of comparison, an ensemble of models is built in all considered comparison methods, using the same number K of models and the same number of trials for setting the model hyperparameters.

6.2. Hyperparameters setting

All methods considered in the ablation study (proposed method, PILSTM and LSTM) employ the architecture reported in Table 1. Note that the number of hidden states of the LSTM layer is equal to the number of outputs, $\mathbf{h}_t$, of the last LSTM layer. Then, the vector $\mathbf{h}_t$ is

Table 8
Performance of RUL predictions of the ablation study (battery case). The values in bold indicate the best performance among the methods.

Metrics	gPILSTM (proposed)	PILSTM	LSTM
CRA	**0.787±0.018**	0.683±0.038	0.740±0.029
AAL	**0.845±0.020**	0.8196±0.227	0.837±0.016
RMSE	**32.276±2.357**	36.474±3.630	36.707±3.076
ASI	**2.620±0.390**	2.641±0.263	3.475±0.507
ADI	**2.779±0.419**	2.843±0.273	3.627±0.515
G	**0.841±0.027**	0.728±0.059	0.762±0.051

concatenated with t, $[\mathbf{h}_t, t]$, and fed into a 3-layer FC NN to perform the RUL regression task. This architecture has been preferred to more complex architectures that have shown limited accuracy improvements at the cost of remarkable increases of the computation burden of model training.

The mini-batch size is set equal to 16 to achieve a robust convergence of the training algorithm using the Adam optimizer [47,48]. The learning rate cyclically varies following the function $f_{lr}(\cdot)$ shown in Fig. 5. This training strategy has been shown to accelerate the convergence and provide satisfactory performance, without requiring manual tuning of the learning rate [49].

Table 2.

Concerning the state-of-the-art methods, PILSTM-Tol uses the same architecture and training hyperparameters as those in ablation study. Regarding BiLSTM, the training dataset is reorganized in time windows of length 5-time steps due to its bidirectional nature. The batch size is set to 32 in the IGBT case study and 512 in the Battery case study, which is characterized by longer run-to-failure trajectories. LSTM-Atten integrates an attention mechanism [50] with a standard LSTM to enhance its capability to capture critical temporal dependencies within the input. Specifically, a multi-head attention layer, appended after the LSTM layers, refines the extracted features through attention-based weighting, enabling the model to adaptively prioritize important temporal patterns and, therefore, improve performance. The number of recurrent layers of the ESN is set to one for the IGBT case study and three for the Battery case study. This choice accounts for the larger number of input signals to process in the latter case study, which requires a deeper architecture to capture hidden features and achieve optimal performance. The other architectures and training parameters are set as recommended in [51]. Regarding CNN, Tables 3 and 4 reports the architecture and hyperparameters, which have been determined by grid search over the number of convolutional layers $(1, 2, 3)$ and fully connected (FC) layers $(1, 2, 3, 4)$. Max pooling layers are not used in the IGBT case study to prevent overfitting given the limited amount of data. The datasets are reorganized in time windows of length 5-time steps. In the KF method, the variance of the measurement noise is set equal to the Mean Square Error (MSE) of the RUL prediction, estimated on a validation set.

The hyperparameters μ, σ and β used in the physics-informed methods to weigh the terms of the loss function (Eq. (20)) were determined through a grid-search. The search domain comprised the values detailed in Table 5. This optimization aimed to minimize the performance metric reported in Eq. (21), wherein the internal weighting coefficients w_m were fixed at 1. This choice gives an overall slightly larger importance to accuracy than stability, since three metrics are assessing accuracy and two are assessing stability.

6.3. IGBT case study

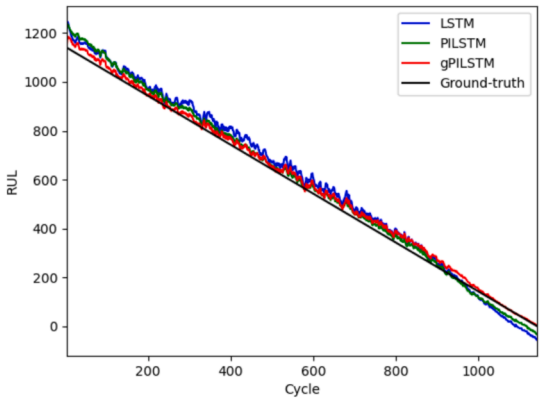
The Insulated Gate Bipolar Transistor (IGBT) is a critical power device used in various industrial systems, including energy conversion devices, motor control systems and photovoltaic solar power system, due to its ability of efficiently handling large voltages and currents [52]. The data considered in this work refer to Accelerated Aging Tests (AATs) of IGBTs operating in inverters of Fully Electric Vehicles [53]. In AATs, the IGBT degradation is obtained by performing predefined and controlled thermal cycles. Given the cost and time required to perform a number of AATs sufficient to develop a prognostic model, a simulator, based on the Miner rule to reproduce the effect of the component degradation on the IGBT electrical resistance [54] and an electrical model of the inverter [55], is employed to simulate run-to-failure degradation trajectories. The parameters of the simulator are set according to the specifications of IGBT (NGTB30N120L2WG) and experts knowledge [56]. Uncertainties related to variations of the external temperature, measurements error, the stochasticity of the degradation process, and errors of the control system during AATs are considered during the simulation. The IGBTs are assumed to fail when their junction temperature reaches a preset threshold. More details on the simulation can be found in [53]. The simulated dataset contains 107 run-to-failure IGBT trajectories. The mean value, of the collector-emitter voltage signal, $\overline{V}_{ce}$, during each thermal cycle, which is recognized as the most effective precursor for diagnostics and prognostics of IGBT [57], is the input of the prognostic model. The output is the prediction of the remaining number of thermal cycles before failure, which will be referred to as RUL. Fig. 6 shows examples of $\overline{V}_{ce}$ evolution during run-to-failure trajectories. To verify the performance and robustness of the proposed method, a 5-fold external cross validation loop is applied to the whole simulation dataset. The obtained mean and standard deviation of the performance metrics from the ablation study are reported in Table 6. Then, Table 7 presents the comparative results between the proposed method and the other state-of-the-art methods considered.

The ablation study demonstrates that incorporating both the physical loss term and the gradient-enhanced loss term effectively enhances the stability of RUL predictions, as evidenced by improvements in the *ASI* and *ADI* metrics. Furthermore, the inclusion of the gradient-enhanced term significantly improves prediction accuracy, as reflected by the increase of the *CRA* metric. An example of RUL predictions over the entire run-to-failure trajectory is shown in Fig. 7. Specifically, Fig. 7(a) presents the results of the ablation study, where Figs. 7(b) and 7(c) illustrate the comparison between the proposed gPILSTM and the other state-of-the-art methods.

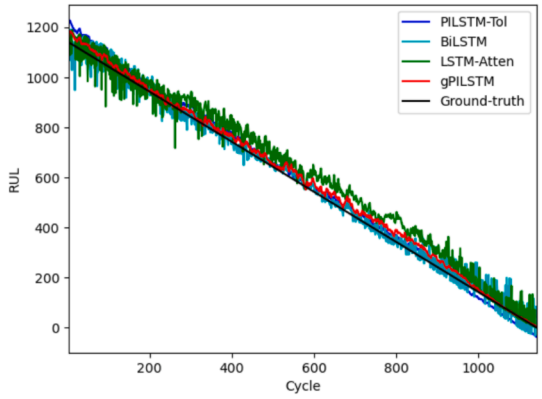
Concerning the other state-of-the-art methods, CNN has the worst performance across all metrics due to the limited number of input signals, which hinders the extraction of representative features from time window data. LSTM-Atten achieves very satisfactory performances on the accuracy metrics, giving CRA value. However, its RUL predictions are more unstable than those of the proposed method, as evidenced by the larger ASI and ADI values. PILSTM-Tol, which is trained using both the data-driven (L_{data}) and physics-informed (L_{phy}) loss functions, yield significantly more stable predictions than BiLSTM and ESN, which are

Table 9
Performance of RUL predictions compared with state-of-the-art methods (battery case). The values in bold indicate the best performance among the methods.

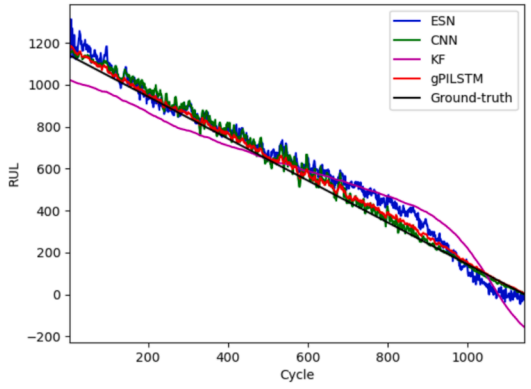
Metrics	gPILSTM (proposed)	PILSTM-Tol	BiLSTM	LSTM-Atten	ESN	CNN	KF
CRA	0.787±0.018	0.707±0.026	0.757±0.013	**0.791±0.012**	0.753±0.021	0.784±0.019	0.695±0.120
AAL	**0.845±0.020**	0.832±0.016	0.767±0.030	0.782±0.033	0.754±0.020	0.687±0.011	0.696±0.174
RMSE	**32.276±2.357**	35.604±2.211	56.241±9.550	50.924±6.684	57.158±10.805	53.906±9.619	83.216±33.896
ASI	2.620±0.390	2.933±0.390	14.318±0.593	14.292±0.916	11.213±2.291	6.128±0.677	**0.715±0.240**
ADI	2.779±0.419	3.084±0.422	18.348±0.678	16.860±0.867	13.582±2.937	6.343±0.633	**0.555±0.154**
G	**0.841±0.027**	0.747±0.046	0.355±0.047	0.419±0.042	0.439±0.079	0.585±0.031	0.600±0.000



(a) comparison within the ablation study.



(b) gPILSTM compared to LSTM based state-of-the-art methods.



(c) gPILSTM compared to non-LSTM based state-of-the-art methods.

Fig. 10. Predictions of the RUL during a run-to-failure trajectory.

Table 10
Performance of the gPILSTM models trained using a decreasing number of run-to-failure trajectories.

Fraction of the total number of run-to-failure trajectories used for training Performance	80 %	70 %	60 %	50 %	40 %
CRA	0.801	0.802	0.790	0.801	0.781
AAL	0.880	0.832	0.847	0.821	0.781
RMSE	31.272	36.588	38.171	40.853	57.463
ASI	2.896	2.842	2.752	3.223	3.289
ADI	3.118	3.116	2.876	3.358	3.356
G	0.836	0.729	0.766	0.423	0

trained using only the data-driven loss function (L_{data}). Both KF and the proposed method provide stable predictions (as indicated by the two best ASI and ADI values) due to the use of Eq. (1). However, KF, which uses a linear model for RUL prediction, is significantly less accurate than the proposed method. Among the different variants of PILSTM, the method that allows obtaining the most stable predictions is the proposed method, based on gPILSTM. This confirms the effectiveness of adding a term related to the gradient of the differential equation $\frac{dRUL}{dt} = -1$ to the loss function. Considering accuracy, the proposed method provides similar results to PILSTM-Tol. Specifically, this latter method slightly overperforms the proposed method when the metrics AAL and RMSE are considered, whereas the proposed method is slightly more accurate when CRA is considered. Note that CRA tends to give more importance to predictions close to the end of life, which is crucial for maintenance decision making. The enhanced performance in CRA of the proposed method can be attributed to the gradient-enhanced term within its loss function, which allows recovering from possibly inaccurate predictions at previous time steps. These inaccuracies are frequently observed as components approach failure, often due to the increased instability of their monitored signals. The overall results indicate that, considering the *G* metric, which provides a tradeoff between accuracy and stability, the proposed method provides the most satisfactory prediction of the IGBTs RUL.

6.4. Lithium-ion battery case

Lithium-ion batteries are used in many industrial applications, such as in the energy, transportation and aerospace industries, due to their high energy density and low self-discharge rate [58]. In this work, we consider the estimation of the RUL of Lithium-ion batteries realistically working under varying operating conditions. The benchmark dataset of MIT and Stanford University [23] is used, as it contains signals measured during run-to-failure battery aging tests performed in three batches of experiments, each one under different fast-charging conditions. The failure criteria is defined as the cycle at which the ratio between the current and the nominal capacity, which is typically considered as the battery State of Health (SoH) indicator, reaches 80 % [59]. Excluding data collected from 21 batteries, which are characterized by extremely short or long lifespan, 103 run-to-failure trajectories are considered. Fig. 8 shows the evolution of the discharge capacity of the 103 batteries of the dataset.

Feature extraction is performed before model development. It transforms the multidimensional time series of the raw signals measured during a cycle into a vector of lumped features. Specifically, 23 knowledge- and statistical-based features, taken from [60–62], are extracted from the data collected when the battery is performing a single cycle. Fig. 9 shows an example of evolution of the extracted features during the life of a battery. Then, 5-fold cross-validation is performed using run-to-failure trajectories from the three batches of experiments. Since the data partition in the 5 folds is done independently from the batch, this cross-validation procedure does not assess the robustness of the models to unseen test conditions. This aspect will be addressed in future work exploring the combination of domain adaptation methods [63] and PINNs.

The obtained mean and standard deviation of the performance metrics from the ablation study are reported in Table 8. Then, Table 9 presents the comparative results between the proposed method and the other state-of-the-art methods considered.

Also in this case, the ablation study demonstrates that the incorporation of the physical loss term significantly enhances the stability of RUL predictions. Notably, unlike the previous case study, the addition of the gradient-enhanced loss term here contributes to both improved stability and increased prediction accuracy. An example of RUL predictions over the entire run-to-failure trajectory is shown in Fig. 10, wherein Fig. 10(a) shows the comparison of the ablation study, Fig. 10

Table 11
Computational time (seconds) of the proposed method and the comparison methods in the lithium-ion battery case study. The experiments are performed on a hardware with GeForce RTX 2070 GPU and Intel Core i5 CPU at 2.40 GHz and 16 G RAM.

	LSTM	BiLSTM	LSTM-Atten	ESN	CNN	PILSTM	PILSTM-Tol	gPILSTM
Training	393.935	97.542	465.623	32.107	92.744	724.39	725.951	758.916
Test (one RUL prediction)	4.60×10^{-4}	9.19×10^{-4}	4.95×10^{-4}	9.07×10^{-4}	9.18×10^{-4}	5.48×10^{-4}	3.27×10^{-4}	3.15×10^{-4}

(b) and (c) show the comparison between the proposed gPILSTM and the other state-of-the-art methods.

Concerning the comparison methods, KF provides the most stable predictions, but does so at the cost of unsatisfactory accuracy. Note that the three physics-informed models (PILSTM, PILSTM-Tol and gPILSTM) provide more stable RUL predictions (smaller ASI and ADI) than all other methods except KF, which confirms the effectiveness of the physical loss function L_{phy}. It can also be seen that PILSTM-Tol and gPILSTM provide the most satisfactory trade-off between accuracy and stability. Specifically, gPILSTM combines stable and accurate predictions, with the best values of the five metrics. Interestingly, CNN and LSTM-Atten exhibit performances comparable to the proposed method with respect to the CRA metric, which evaluates relative errors. This is because they provide accurate RUL predictions near the end of the component life, where even small absolute errors can significantly decrease the CRA. In this phase, typically characterized by large variations of the measured signals, the short-term focus of CNN and the capability of the attention mechanism of reweighting features are shown to be more effective for accurate RUL prediction than the long-term memory of traditional LSTM-based models.

To verify the performance of the proposed gPILSTM in scenarios where the number of run-to-failure data is limited, we have conducted the following analysis considering various gPILSTM models trained on datasets containing a decreasing number of run-to-failure trajectories. The results in Table 10 show that the performance remains satisfactory until the training data are reduced to 40 % of the available amount, when, the diversity of the training data becomes insufficient to allow representing the distribution of the test data.

Latency is an inevitable issue in the development and application of data-driven models for fault prognostics. The computational times needed for model training and test in this work are reported in Table 11, considering the same amount of training and testing data. The computational effort necessary to train LSTM-based models is larger than that of CNN due to their sequential nature. The ESN model requires significantly less time for training since it does not rely on an iterative training process. However, testing the ESN model demands more computational effort due to the complexity of reservoir computing. Among the LSTM-based models, the training of gPILSTM is the most time consuming due to the presence of the additional terms in the loss function. Similar considerations apply to the time needed to compute the RUL of a component during the test phase. Considering a hardware with GeForce RTX 2070 GPU and Intel Core i5 CPU at 2.40 GHz and 16 G RAM, the proposed method can be trained in $<$15 min and is able to provide a RUL prediction for a test component with a latency of 3.15×10^{-4} s, which is acceptable for real applications in the electronic industry.

7. Conclusions

The problem of obtaining RUL predictions not only accurate but also stable has been addressed in this work. The aim is to improve the trustworthiness of the prognostic model by providing RUL predictions that do not rapidly change in small intervals of time, to avoid misleading maintenance decision makers. The proposed method is based on Gradient-enhanced Physics-informed LSTM. LSTM cells have been chosen because of the sequential nature of sensor measurements and the nonlinear behaviour of the degradation process. The use of PINN allows leveraging in the model training the knowledge that the ground-truth RUL is reduced by one time unit for every unit of time elapsed, which is formulated as $\frac{dRUL}{dt} = -1$. A term related to the gradient of the physical law is added to the loss function to allow recovering from inaccurate RUL predictions that typically tend to occur at the end of the component life. An ensemble strategy is proposed to properly set the weights associated to the terms of the loss function when a limited number of run-to-failure degradation trajectories is available for model development. The proposed method has been verified considering two prognostic case studies which refer to IGBTs and Lithium-ion batteries. The obtained results show the superior performance of the proposed gPILSTM in terms of both accuracy and stability when compared to other state-of-the-art methods.

On the other hand, the applicability of the proposed method depends on the amount of data available to train the model and set its hyperparameters, which may be not sufficient in some industrial applications. Furthermore, the use of fixed weights in the loss function could reduce the performance of the trained model. Future work will also address these practical issues by: (1) enriching the training dataset using generative models (e.g. diffusion models); (2) leveraging transfer learning and domain adaptation techniques to allow using data collected from other components and under different operating conditions; (3) developing methods for adaptively and automatically setting the weights of the loss functions. These advancements will be carried out systematically, and the accuracy and stability of the resulting RUL predictions will continue to be evaluated also by comparison with those of other prognostic methods proposed in the literature.

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CRediT authorship contribution statement

Chenyang Lai: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Data curation, Conceptualization. **Piero Baraldi:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization. **Enrico Zio:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

1. Accuracy Metrics

(1) *CRA*. Considering the generic *r*th run-to-failure trajectory, *Relative Accuracy*, RA_t^r, at time t is defined as:

$$RA_t^r = 1 - \frac{\left|\widehat{RUL}_t^{\,r} - RUL_t^r\right|}{RUL_t^r} \tag{24}$$

$\widehat{RUL}_t^{\,r}$ is the predicted RUL at time t of test component r and RUL_t^r is the corresponding ground truth. *CRA* is the average of *RA* over all the time instants and all the trajectories:

$$CRA = \frac{1}{R^{test}} \sum_{r=1}^{R^{test}} \frac{1}{t_f^r} \sum_{t=1}^{t_f^r} RA_t^r \tag{25}$$

t_f^r is the ground truth failure time of test component r. Larger the value of *CRA* is, more satisfactory the performance is.

(2) *RMSE*. *RMSE* is a commonly used statistical measure of the average magnitude of error between predicted and actual RUL:

$$RMSE = \frac{\sqrt{\sum_{r=1}^{R^{test}} \sum_{t=1}^{t_f^r} \left(RUL_t^r - \widehat{RUL}_t^{\,r}\right)^2}}{\sum_{r=1}^{R^{test}} t_f^r} \tag{26}$$

Smaller is the *RMSE*, more accurate are the RUL predictions.

(3) *AAL*. The AAL investigates whether the RUL prediction falls within a specified interval containing the ground truth RUL. The $\alpha - \lambda$ *Accuracy* (*AL*) is a binary metric that evaluates whether the RUL prediction at a specific time instant t falls within a specified α-bound of the ground truth RUL. α-bounds are expressed as percentage of the ground truth RUL:

$$AL_t^r = \begin{cases} 1 \text{ if } (1-\alpha)\cdot RUL_t^r \le \widehat{RUL}_t^{\,r} \le (1+\alpha)\cdot RUL_t^r \\ 0 \text{ otherwise} \end{cases} \tag{27}$$

Then, *AAL* is obtained by computing the average AL_t^r of each time instant, $t = 1 \ldots t_f^r$, of each trajectory, $r = 1, \ldots, R^{test}$:

$$AAL = \frac{1}{R^{test}} \sum_{r=1}^{R^{test}} \frac{1}{t_f^r} \sum_{t=1}^{t_f^r} AL_t^r \tag{28}$$

The larger is the *AAL* value, the more accurate is the model, since more predictions fall within the corresponding α-bounds.

2. LSTM for RUL prediction

LSTM is a type of Recurrent Neural Networks for learning tasks involving the processing of time series [41]. LSTMs control the information flow using input, forget and output gates, which allow them remembering, when needed, information for long periods of time. Specifically, at time t, the LSTM network receives in input the L-dimensional vector of measurements, $\mathbf{x}_t^{test} = \left[x_t^1, x_t^2, \cdots, x_t^L\right] \in \mathbb{R}^{L\times 1}$ and provides in output the prediction $\widehat{RUL}_t$. The architecture of a LSTM model is composed of multiple layers of LSTM cells, which are, then, followed by fully-connected (FC) layers. The LSTM cells process the temporal behaviour of the degradation and provide in output the vector of features $\mathbf{h}_t$, whereas the FC layers map the extracted features $\mathbf{h}_t$ into the RUL. Sensor measurements $\mathbf{x}_t^{test}$ are used as input only to the first LSTM layer. In this work, following the approach proposed in [36] to improve the accuracy of the RUL predictions, time t is used as an explicit input of the FC layers together with the output of the LSTM cells. The unfolded LSTM structure is shown in Fig. 11. Each layer of LSTM, which contains a chain of repeating cells, is defined by cell structure and number of nodes within the cell. Repeating cells within LSTM layer have the same structure and parameter values.

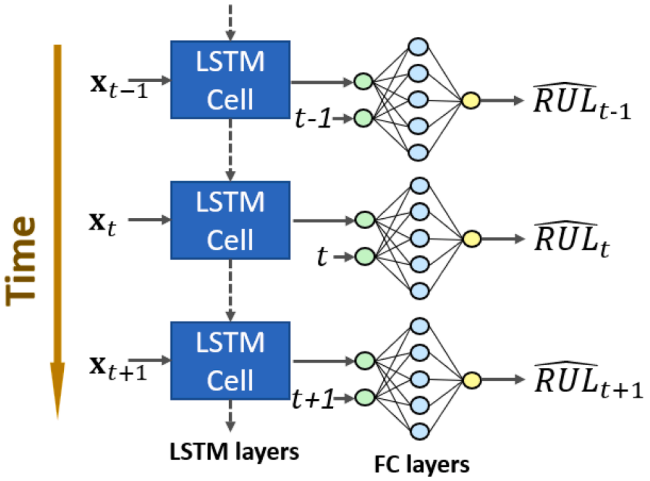


Fig. 11. Architecture of LSTM network.

LSTM cell structure at time t is shown in Fig. 12. We differentiate output and states of LSTM cell denoted as $\mathbf{h}_t$ and $\boldsymbol{c}_t$, respectively. Vector size of the output and states is the same and it is defined by number of nodes in the cell. Let denote m as number of nodes, so $\mathbf{h}_t \in \mathbb{R}^{m\times 1}$ and $\boldsymbol{c}_t \in \mathbb{R}^{m\times 1}$. The $\mathbf{h}_{t-1}$ and $\boldsymbol{c}_{t-1}$ of LSTM cell at time $t-1$ will serve as an input to LSTM cell at time t, whereas the other input is $\mathbf{x}_t$. There are three gates that control the information flow within cell: (1) input gate $\mathbf{i}_t \in \mathbb{R}^{m\times 1}$ controls what information based on output $\mathbf{h}_{t-1}$ and $\mathbf{x}_t$ will be passed to memory cell, (2) output gate $\mathbf{o}_t \in \mathbb{R}^{m\times 1}$ controls what information will be carried to the next time step and (3) forget gate $\mathbf{f}_t \in \mathbb{R}^{m\times 1}$ controls how memory cell will be updated. All LSTM cells that are used in the models are implemented as follows:

$$\mathbf{i}_t = \sigma(W_i\mathbf{x}_t + U_i\mathbf{h}_{t-1} + \mathbf{b}_i) \tag{29}$$

$$\mathbf{o}_t = \sigma(W_o\mathbf{x}_t + U_o\mathbf{h}_{t-1} + \mathbf{b}_o) \tag{30}$$

$$\mathbf{f}_t = \sigma\left(W_f\mathbf{x}_t + U_f\mathbf{h}_{t-1} + \mathbf{b}_f\right) \tag{31}$$

$$\mathbf{a}_t = \tanh\left(W_c\mathbf{x}_t + U_c\mathbf{h}_{t-1} + \mathbf{b}_c\right) \tag{32}$$

$$\mathbf{c}_t = \mathbf{f}_t \circ \mathbf{c}_{t-1} + \mathbf{i}_t \circ \mathbf{a}_t \tag{33}$$

$$\mathbf{h}_t = \mathbf{o}_t \circ \tanh\left(\mathbf{c}_t\right) \tag{34}$$

where variable weights and bias to be computed during training process are W_i, W_o, W_f, $W_c \in \mathbb{R}^{m\times L}$, U_i, U_o, U_f, $U_c \in \mathbb{R}^{m\times m}$, $\mathbf{b}_i$, $\mathbf{b}_o$, $\mathbf{b}_f$, $\mathbf{b}_c \in \mathbb{R}^{m\times 1}$. $\circ$ is element-wise multiplication of two vectors (Hadamard product). σ is element-wise logistic sigmoid activation function. Connection between different layers of LSTMs is achieved such that the output of former layer is as an input to the next layer.

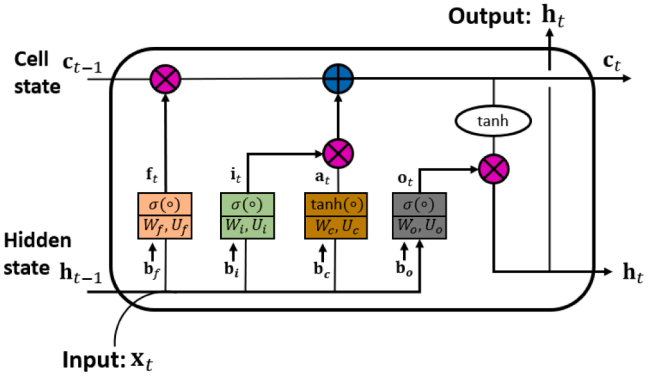


Fig. 12. Single LSTM cell structure.

Data availability

Data will be made available on request.

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